

Battery Dispatch under Forecast Uncertainty

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Abstract

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Nomenclature

Abbreviations

MPC	model predictive control
PV	photovoltaic
RL	reinforcement learning
SoC	state of charge
SESOI	smallest effect size of interest
P1–P4	rule-based, point-forecast MPC, probabilistic-forecast MPC, RL policy
P2m, P2n	back-off MPC, fitted-model MPC
P3s, P3r	scenario-based and robust MPC

Symbols

t, H	hour, MPC horizon
L_t, PV_t, n_t	load, PV output, net load $L_t - PV_t$
b_t, g_t	battery power (charging positive), grid exchange (import positive)
p_t	day-ahead price
$\bar{P}, \bar{E}$	subscribed import power, export cap
ρ	overrun penalty above $\bar{P}$
m	import back-off margin of P2m
$D(b)$	degradation cost of a battery trajectory
$C_\pi(d)$	daily cost of policy π on day d
R_π	regret of policy π against the perfect-foresight oracle
V_π	share of the perfect-foresight value captured by policy π
λ	forecast-error scaling (skill)
β	weight of the CVaR term in scenario MPC

1. Introduction

A battery behind the meter of a commercial prosumer earns its value one decision at a time. Charging now commits energy on the basis of today’s forecast of load, photovoltaic (PV) output and prices; holding keeps the option to act on a better forecast later. Model predictive control (MPC) resolves this trade-off by re-planning every hour on the latest forecast, and the quality of its decisions therefore depends on what the forecast tells it. This paper asks what forecast information is worth to that decision, and where its value comes from. We separate three kinds of information: the *uncertainty*

around a point forecast, the *structure* that a model-based controller brings compared with a model-free learner, and the *skill* of the forecast itself.

The question matters because each kind of information has a cost. A probabilistic forecast needs an error model and a scenario generator, and a stochastic or robust controller needs more computation than a deterministic one; a reinforcement learning (RL) agent needs no battery model but a long training history; a more skilful forecast needs better data and models. An operator who must choose where to spend needs to know which of these pays, under which tariff, and by how much.

Stochastic MPC on probabilistic forecasts has been reported to outperform MPC on point forecasts for residential PV–battery systems and home energy management [1, 2, 3], and for stationary batteries facing demand charges [cite Kumar et al. 2018 and 2019: bibliography exception requested, NOTES.md]. The direction of that result is therefore known. Three things are less settled. First, the simplest alternative to a distribution is a safety margin: plan on the point forecast, but keep the import below the contracted limit minus a tuned back-off. This practice has been described as ad hoc [cite Kumar et al. 2019: bibliography exception requested], and guard bands of this kind have been proposed for price arbitrage [4], but we have not found a study that tunes such a margin on validation data and compares it out of sample with stochastic MPC on the same point forecasts. Second, the source of the value is rarely separated. When the tariff is affine in the grid exchange and no constraint depends on load or PV, the PV and load forecasts cannot change a risk-neutral controller’s decision; any value they have must come from the tariff’s non-linearities, such as a penalty on import above a subscribed power. The mechanism is elementary and known in stochastic programming, but its share of the value of probabilistic forecasts for a behind-the-meter battery has, to our knowledge, not been measured on held-out data. Third, comparisons of RL and MPC for batteries and buildings reach mixed conclusions [5, 6, 7]; in the closest precedent, a set of multi-energy system case studies [5], the MPC’s model error and its forecast error are not varied separately, so the conditions under which a model-free agent can win are not isolated.

We study one grid-connected commercial prosumer in France, with load, rooftop PV, a battery and a connection point with a subscribed import power and an overrun penalty, priced at the day-ahead market (Section 3). Four families of policies share the same point forecasts: a rule-based policy, MPC on point forecasts, MPC on probabilistic forecasts built from those point

forecasts (scenario-based or robust), and a model-free RL agent (Section 4). Every policy is tuned on a validation year and evaluated on held-out test years that are opened once, against a perfect-foresight oracle, by the share of the oracle’s value that it captures. The hypotheses, tests, smallest effect of interest and reading rules were fixed in a written protocol before any data from the test years were requested (Section 5). The study therefore tests the following claims; each is stated as a hypothesis, and the results report whether it is supported, weakened, refuted or inconclusive at the available power.

1. **The tariff creates the value (C4).** We test whether the gain of scenario MPC on probabilistic forecasts over MPC on point forecasts is created by the overrun penalty: (a) with the penalty removed, the difference in value captured lies within the band set by the smallest effect of interest; (b, secondary) the difference is larger at the main penalty level than with no penalty. The analytical anchor is Proposition P1 (Section 3.2): with an affine tariff, no constraint that depends on PV or load, a nonempty feasible set, a matched plant and the planned first action applied, PV and load forecasts, point or probabilistic, do not change the optimal decisions of point MPC or of risk-neutral scenario MPC. This contribution is a quantified decomposition of known mechanisms, not a new mechanism.
2. **Probabilistic forecasts against a tuned safety margin (C1b).** We test whether MPC on probabilistic forecasts captures more of the perfect-foresight value on held-out years than point-forecast MPC with a validation-tuned import back-off, with the same point forecasts and horizon, by at least the smallest effect of interest.
3. **The value of probabilistic forecasts (C1a).** We test whether MPC on probabilistic forecasts captures more of the perfect-foresight value than MPC on the same point forecasts with the same horizon, by at least the smallest effect of interest, and whether most of the gain comes from avoided overrun penalties rather than from energy arbitrage. Because the point forecast is held fixed, the difference isolates the value of the uncertainty information.
4. **The value of forecast skill (C3).** Scaling the operational forecast error from zero (perfect foresight) upwards, we test whether the gain of probabilistic over point forecasts grows with the size of the forecast error over the grid [skill-scaling grid, from generated/] and vanishes at

perfect foresight. We also estimate the skill equivalent of the probabilistic forecast: the error reduction that point-forecast MPC would need to capture the same value.

5. **Model-based control versus model-free learning (C2).** We test whether an RL agent captures no more value than point-forecast MPC when the MPC’s battery model is correct (C2a); whether a mismatch between the battery and the MPC’s model narrows the gap between RL and MPC (C2b, a difference in differences); and, as a secondary test, whether RL keeps any advantage on the mismatched plant over MPC whose battery model is fitted on validation data (C2c).

The order of the list is the order of the contributions: C4 and C1b are what this study adds to the literature, C1a and C3 place known directions on a pre-registered, held-out footing, and C2a replicates a result reported in several settings.

Section 2 positions the study against the literature. Section 3 defines the site, the dispatch problem and the metrics, Section 4 the forecasts and policies, and Section 5 the data, the periods and the pre-registered tests. Section 6 reports the outcomes, Section 7 discusses them, Section 8 states the limits of the study and Section 9 concludes.

2. Related work

Stochastic MPC and the value of probabilistic forecasts for batteries. Scenario-based stochastic MPC for PV–battery systems uses probabilistic forecasts of PV and load to plan a first action that is good across scenarios. Van der Meer et al. [1] generate autocorrelated scenarios from probabilistic forecasts for a residential battery and propose a first-step decision common to all scenarios; our scenario MPC uses the same non-anticipativity structure (Definition S1). Groß et al. [3] combine a stochastic forecast model of household load and PV with nonlinear converter and standby losses, and validate the controller in a field test. Blonsky et al. [2] compare heuristic, deterministic and stochastic control for home energy management under realistic uncertainty. For stationary batteries facing demand charges, stochastic MPC was proposed and benchmarked against deterministic MPC [cite Kumar et al. 2018 and 2019: bibliography exception requested, NOTES.md]; this is the closest precedent for C1a and for its penalty mechanism. Robust MPC with neural-network forecasts has been used for peak shaving on a university campus [8], and stochastic programming with probabilistic time-series forecasts for peak shaving in

buildings [9]. The direction of C1a is therefore established. Relative to these studies, the present one holds the point forecast fixed, so that the difference between point and probabilistic MPC isolates the uncertainty information, and it evaluates both on held-out years with pre-registered tests.

Safety margins and peak or capacity tariffs. A cheaper way to hedge forecast error is to keep a deterministic controller and tighten its constraints. In the stationary-battery benchmark cited above, such a constraint back-off improved deterministic MPC but was described as ad hoc [cite Kumar et al. 2019]. Matsumoto and Sasanuma [4] build conservative guard bands from rolling quantiles of price-forecast errors and embed them in a deterministic arbitrage scheduler; in a stylized model the optimal buffer grows with forecast noise. Imannezhad et al. [10] adjust price signals with quantile bounds of past forecast errors and report the share of the perfect-foresight arbitrage value that their scheduler captures, the metric used here. Both works apply the margin to prices in energy arbitrage. The margin tested in C1b applies to the import limit of a prosumer, is tuned on a validation year and is compared out of sample with stochastic MPC on the same point forecasts. The tariff matters as much as the controller. Under a tiered peak-power tariff, Pérez-Piñero et al. [11] find that MPC with simple forecasts comes close to the prescient bound and saves considerably more than the best rule-based policy they consider; this is the strongest reason to expect that point-forecast MPC, with or without a margin, may leave little for probabilistic forecasts to add. Subscribed-capacity tariffs of the kind our site faces have been studied with perfect-information optimization [cite Berglund et al. 2019: bibliography exception requested], and capacity-based tariff designs are compared for their peak-shaving effect by Hofmann et al. [12].

Where the value comes from. When uncertainty enters only the constant term of a stochastic program’s cost, the value of the stochastic solution is zero [cite Birge 1982 and the Birge–Louveaux textbook: bibliography exception requested]. Proposition P1 is this observation for the dispatch problem of Definition S1: with an affine tariff and no constraint that depends on PV or load, PV and load forecasts do not change the decisions of point MPC or risk-neutral scenario MPC. C4 uses it as a null anchor and measures, on held-out years, how much of the gain of probabilistic forecasts is created by the overrun penalty. It is a quantified decomposition of a known mechanism, not a new mechanism.

Reinforcement learning versus MPC. Model-free RL for energy systems has been reviewed broadly [13, 14]. Arroyo et al. [15] contrast RL and MPC for building energy management in constraint satisfaction, computation, adaptability and intelligibility, and merge the two; hybrids of this kind are outside the scope of C2. Ceusters et al. [5] find, in multi-energy system case studies, that a model-free agent can match or outperform a linear MPC that relies on imperfect forecasts; model error and forecast error act together there. In a laboratory validation on four replicas of residential houses, Ruddick et al. [6] report similar costs for rule-based control, decision trees and MPC, with safe RL behind. Langer and Volling [16] compare a deep RL agent for a home with a heat pump, PV and a battery with a full-information MPC bound and a rule-based benchmark. Further comparisons of RL with MPC or mixed-integer programming exist for building thermal systems [7, 17] and for power plant scheduling [18]. For battery dispatch specifically, Sage and Zhao [19] show that design choices of the RL experiment, such as time features in the state, matter; our agent’s observation encodes hour and weekday cyclically, as they recommend (Section 4.2). C2a is therefore a replication in a crowded literature. What C2b adds is a controlled contrast: the same forecasts and site with a matched and a mismatched battery, so that the effect of model error on the RL–MPC gap is estimated as a difference in differences; C2c asks whether a cheap fix of the MPC’s model, a power-dependent efficiency fitted on validation data [cite Gonzalez-Castellanos et al. 2020: bibliography exception requested], removes any advantage RL gains from the mismatch.

Forecast skill and the value of forecasts for decisions. Forecast accuracy and decision value are not the same thing. Decision-focused and value-oriented learning train the forecaster for the downstream decision [20, 21, 22], including for PV–battery scheduling [23]. The present study does the opposite: it holds the forecaster fixed and measures the value of its information as a function of the size of its error, keeping the operational error pattern. The influence of solar forecast accuracy on the revenue of PV and battery plants has been assessed [24]; that is the value of point skill, whereas C3 asks how the value of uncertainty information changes with skill, and expresses it as a skill equivalent. The day-ahead price forecaster follows the open benchmark and best practices of Lago et al. [25].

Benchmarks and environments. Open benchmarks exist for energy management: EMSx provides real microgrid sites with PV, storage and forecasts and

compares MPC with other stochastic control methods [26]; CityLearn [27] and Gym-ANM [28] provide RL environments for building communities and distribution networks. They do not provide a French commercial prosumer with day-ahead prices and a subscribed-power penalty at a single connection point, which is the setting of this study (Section 5.1).

Position of this study. Relative to this literature, the study makes two contributions and supports three known directions with a stricter design. The contributions are the decomposition of the value of probabilistic forecasts into the channel created by the overrun penalty and the rest (C4), and the out-of-sample comparison of probabilistic forecasts with a validation-tuned safety margin (C1b). The known directions are the value of probabilistic over point forecasts (C1a), its growth with forecast error (C3) and the comparison of RL with MPC (C2), each tested here on held-out years, with the point forecasts shared across policies and the tests fixed before any test data were seen.

3. Site, dispatch problem and metrics

3.1. Site and tariff

3.2. Dispatch problem

3.3. Cost, oracle and value captured

4. Forecasts and dispatch policies

4.1. Point and probabilistic forecasts

4.2. Policies

4.3. Plant variants

5. Experimental design

5.1. Data and periods

All inputs are public time series; the site built from them is described in Section 3.1.

Day-ahead prices. Day-ahead prices for the French (FR) bidding zone, published under Article 12(1)(d) of Commission Regulation (EU) No 543/2013, were obtained from the ENTSO-E Transparency Platform (<https://transparency.entsoe.eu/>), retrieved on [retrieval date, from the data vintage record]. ENTSO-E does not endorse this work. Prices for a delivery day are published on the day before, after the market clears, so the controller knows them from that moment; only prices beyond the published horizon are forecast (Section 4.1). Where the market clears at a sub-hourly resolution (from [date of the change to sub-hourly clearing, confirmed by the lab]), prices are converted to their hourly mean before any use, in every period. The platform’s terms of use allow the prices to be used for research and results derived from them to be published with the platform named as the source; they do not allow the price series to be redistributed. The paper therefore reports only derived results (costs, value captured, test statistics and figures of policy results), and no table or figure reproduces the price series. The replication package contains the download code and, for every file retrieved, the request without the access token, the retrieval time and a cryptographic hash; a replicator registers for an own access token, downloads the same ranges and checks them against the hashes. Because published data can be revised, a later download may differ from ours; the check reports which ranges differ.

Load. The load of the site is the national consumption profile published by the French transmission system operator RTE (éCO2mix), scaled to the annual energy and peak of a commercial site (Section 3.1). Its point forecast is RTE’s own day-ahead consumption forecast, with the same scaling, used only from the time at which it is available to the controller.

Weather and PV. Realized PV output is computed from Open-Meteo historical weather (irradiance components and temperature) through a PV model with fixed losses [cite the PV modelling library and the Open-Meteo dataset record, once verified]. The PV point forecast applies the same model to archived numerical weather forecasts issued on the day before, not to re-analysis data; the archive’s coverage by period and lead time is recorded in the data manifest and reported with the results.

Periods. The data are split once into a training period ([training period, from generated/]), used for the forecast-error models, RL training and rule thresholds; a validation year ([validation year, from generated/]), used for every tuning and selection decision; and held-out test years ([test period and its

number of days, from generated/]), used only for the outcomes. By the protocol, the test data are requested only after every tuning and selection decision has been fixed, and they are opened once. The price forecaster’s rolling calibration window may reach back before the training period, using past data only at each issue time.

5.2. Hypotheses and tests

5.3. Inference and power

6. Results

6.1. Where the value of forecast information comes from

6.2. Probabilistic forecasts against a tuned safety margin

6.3. Value of probabilistic forecasts

6.4. Value of forecast skill

6.5. Reinforcement learning against model predictive control

7. Discussion

8. Limitations

9. Conclusion

Data availability

The code, the experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author’s project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). The input data are public (day-ahead prices, national electricity data and weather data); their sources and access dates are given in Section 5.1.

CRediT authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Software, Investigation, Validation, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

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During the preparation of this work the author used large-language-model assistants (Claude, Anthropic) to draft and revise parts of the text and to write and review parts of the analysis code. After using these tools, the author reviewed, checked and edited the content as needed and takes full responsibility for the content of the published article.

Appendix A. Proof of Proposition P1

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